

ALWAYS DRAW THE PICTURE FIRST

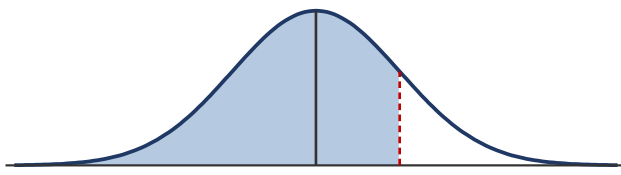
Bell → mark μ → mark x → shade what is asked

1. Draw the bell and put μ under the peak.
2. Mark your x and draw a vertical line up to the curve.
3. **Shade the region the question describes.**
4. Shaded on the LEFT → NORMDIST straight off. Shaded on the RIGHT → $1 - \text{NORMDIST}$.

The picture tells you whether to subtract from 1. That is the whole trick.

THE FOUR SHAPES YOU WILL DRAW

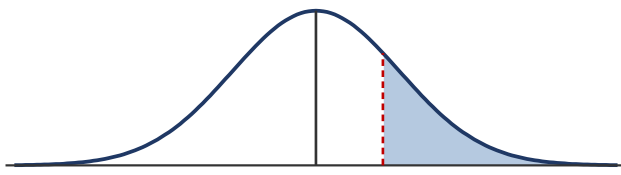
1 · $P(X < x)$ — shade LEFT



read it straight off NORMDIST

$$P(X < x) = \text{NORMDIST}(x, \mu, \sigma, \text{TRUE})$$

2 · $P(X > x)$ — shade RIGHT

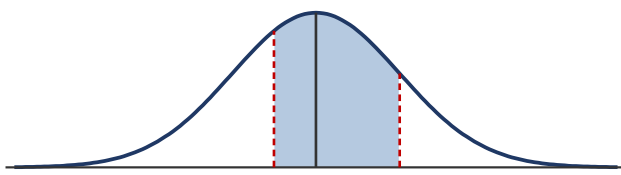


NORMDIST gives the WRONG side — subtract

$$P(X > x) = 1 - \text{NORMDIST}(x, \mu, \sigma, \text{TRUE})$$

THE FOUR SHAPES (continued)

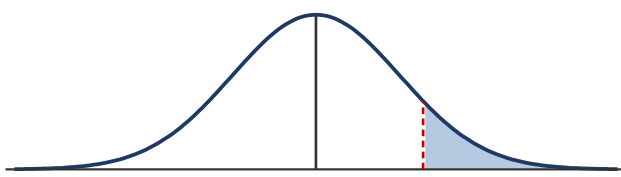
3 · $P(a < X < b)$ — shade the MIDDLE



subtract the two left-areas

$$P(a < X < b) = \text{NORMDIST}(b, \dots) - \text{NORMDIST}(a, \dots)$$

4 · BACKWARDS — the area is known



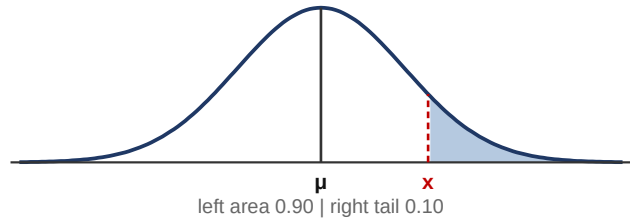
you are given 10%, you want x

$$x = \text{NORMINV}(p, \mu, \sigma)$$

Big area minus small area. Always the bigger cut-off first.

THE TOP-10% TRAP — draw it and you cannot get it wrong

“top 10%” looks like this



The shaded tail is **0.10**, but NORMINV wants the area to its **LEFT**, which is **0.90**.

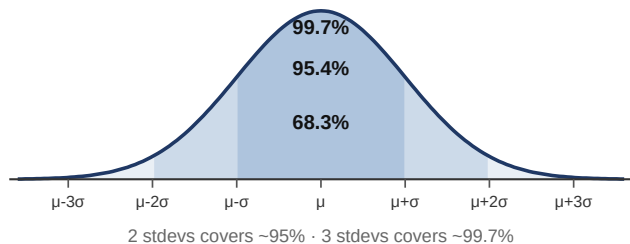
Top 10% → $=\text{NORMINV}(0.90, \mu, \sigma)$

Bottom 5% → $=\text{NORMINV}(0.05, \mu, \sigma)$

Lecture: top 10% of $N(80, 10) = 92.8$ · worst 5% of $N(7, 10) = -9.45$

THE 68 – 95 – 99.7 PICTURE

The 68 - 95 - 99.7 picture

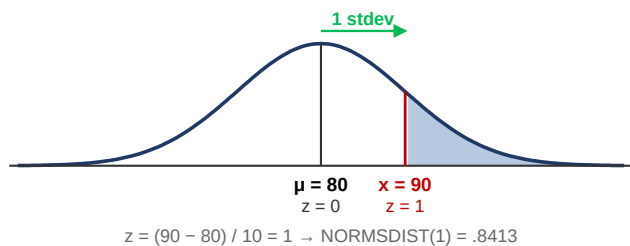


2 stdevs covers ~95% · 3 stdevs covers ~99.7%

Sketch this when the question says “within 2 standard deviations”.
Weight $\mu=180$, $\sigma=10$ → 95.4% of men are between 160 and 200.

WHAT A Z-SCORE ACTUALLY DOES

z just relabels the axis

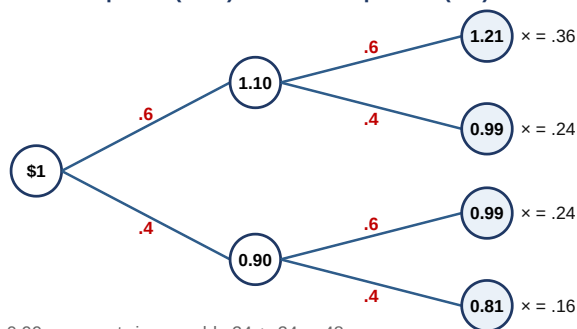


$$z = (x - \mu) / \sigma$$

Same curve, relabelled axis. z just counts stdevs from the middle.

Comparing two things? Draw two curves, mark each value, compare $|z|$. Shanghai $z=1.54$ vs NYSE $z=2.21$ → NYSE was rarer even though its return was smaller.

HOW TO DRAW A PROBABILITY TREE

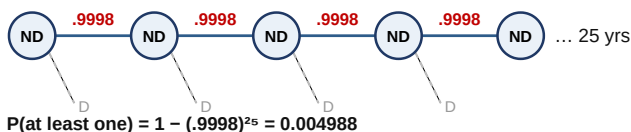
ALONG a path $\times$ (AND) · BETWEEN paths + (OR)0.99 appears twice $\rightarrow$ add: $.24 + .24 = .48$ ALONG a path $\times$ (AND) · BETWEEN paths + (OR)

1. One branch per outcome at each stage; write the probability **on** the branch.
2. Multiply along a path to get that path's probability.
3. Add the paths that give the same answer.
4. Check every path probability sums to 1.

Here 0.99 is reached two ways, so $P(0.99) = .24 + .24 = .48$. Then $E = 1.21(.36) + 0.99(.48) + 0.81(.16) = 1.04$.

“AT LEAST ONE” — draw the NEVER path

Never defaults = one long multiply



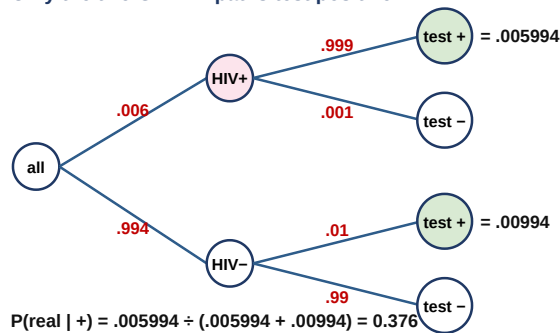
$$P(\text{never}) = (1 - p)^N$$

$$P(\text{at least one}) = 1 - (1 - p)^N$$

Never add up the “at least one” cases. There is only ONE never-path — draw that, then subtract.

BAYES — the tree does the work

Only the two GREEN paths test positive



1. Split the population first: **has it / does not**.
2. Then split each by the **test result**.
3. Circle every path that tests **positive**.
4. Answer = the path you want $\div$ all circled paths.

A 99.9% accurate test for a **rare** thing is still mostly false alarms — the healthy branch is enormous.

OUTCOME TABLES — $E(X)$ is the balance point

$$E(X) = \sum x \cdot P(x)$$

$$V(X) = \sum P(x) \cdot (x - E(X))^2$$

$$\text{stdev} = \sqrt{V(X)}$$

x	P(x)	x·P	P(x-E) ²
0	.1	0	.1(0-1.8) ² =.324
1	.3	.3	.3(1-1.8) ² =.192
2	.3	.6	.3(2-1.8) ² =.12
3	.3	.9	.3(3-1.8) ² =.432
sum	1.0	1.8	0.96

$E = 1.8 \cdot V = 0.96 \cdot \text{stdev} = 0.9797$. Build the two right-hand columns and the answer falls out.

Square the gap **before** weighting by the probability, never after.

IF IT THEN SAYS “profit = $a + b \times X$ ”

$$E(a+bX) = a + b \cdot E(X)$$

$$\text{stdev}(a+bX) = |b| \cdot \text{stdev}(X)$$

a slides the curve sideways. It **never** changes the width.

$Y = -50 + 200X \rightarrow E = 310, \text{stdev} = 195.96$. A tiered bonus is **NOT** a straight line — rebuild the table instead.

TWO VARIABLES — do SAXBY first, THEN draw the curve

Step 1: get the new μ and σ → Step 2: draw ONE bell

$$E(aX+bY) = a \cdot E(X) + b \cdot E(Y)$$

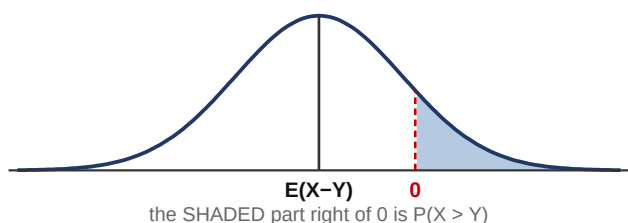
$$V(aX+bY) = a^2V(X) + b^2V(Y) + 2ab \cdot \text{sd}X \cdot \text{sd}Y \cdot \text{Corr}$$

The question asks about	a	b
the total X+Y	1	1
$P(X > Y)$	1	-1
weighted 35 / 65	.35	.65

Once you have the combined μ and σ , it is an ordinary page-1 bell-curve question.

 $P(X > Y)$ — the picture that explains it

draw the curve of $X - Y$, then mark ZERO



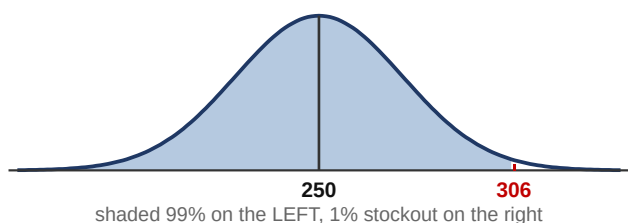
$$P(X > Y) = P(X - Y > 0)$$

1. Set $a=1$, $b=-1$ → you now have the variable $X-Y$.
 2. Its mean is $100-150 = -50$, its stdev is **9.22**.
 3. Draw that bell and mark **0** on it.
 4. Shade to the RIGHT of zero. That shaded bit is the answer.
- The sketch above shows a typical case. In the lecture example zero sits 5.4 stdevs above the mean of -50 , so the shaded sliver is ≈ 0 — walk-ins essentially never beat phone orders.

Mark **ZERO** on the $X-Y$ curve, not the mean. Getting that line in the right place is the whole question.

SERVICE LEVEL / STOCK LEVEL

cover 99% of days



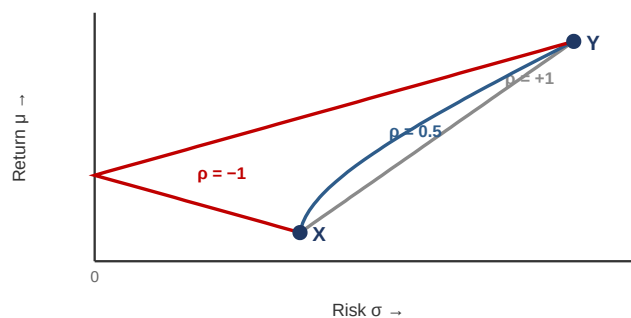
$$\text{level} = \text{NORMINV}(0.99, \mu_{\text{total}}, \sigma_{\text{total}})$$

Combine the two demands with SAXBY first ($a=1$, $b=1$), THEN use NORMINV. $N(250, 23.77) \rightarrow 305.3 \rightarrow$ order **306**.

Always round **UP**. 305.3 units of stock does not exist.

PORTFOLIO — the risk / return plane

Lower correlation bends the curve LEFT



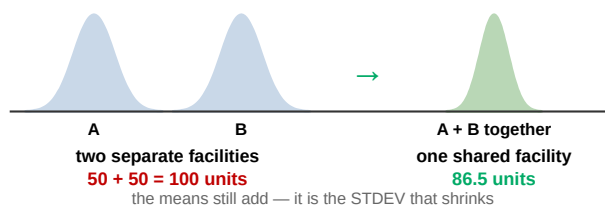
Each point is one weighting. $w=0$ is all Y, $w=1$ is all X.

Corr	What the curve does
+1	straight line — no benefit
0	bends left — real risk reduction
-1	bends to the axis — risk can hit zero

The leftmost point of the curve is the **minimum-risk portfolio**. It can sit further left than BOTH assets — that is the portfolio effect.

POOLING — why sharing needs less

Pooling makes the curve **NARROWER**



separate: $\Sigma(\mu + k\sigma)$ pooled: $\Sigma\mu + k \cdot \sigma(X+Y)$

GPUs: $50 + 50 = 100$ separately, versus 86.5 pooled. The means add either way — it is the **stdev** that shrinks, so less safety cover is needed.

THE 60-SECOND DECISION

What the question hands you	Draw / do
one μ and σ , wants a %	bell + shade → NORMDIST
one μ and σ , wants a value	bell + shade → NORMINV
outcomes with probabilities	the 4-column table
stages, or “at least one”	a tree
a correlation	SAXBY , then one bell
weights / a portfolio	risk-return plane

If you are stuck: **draw the bell and shade it**. Half the marks are in knowing which side of the line you want.